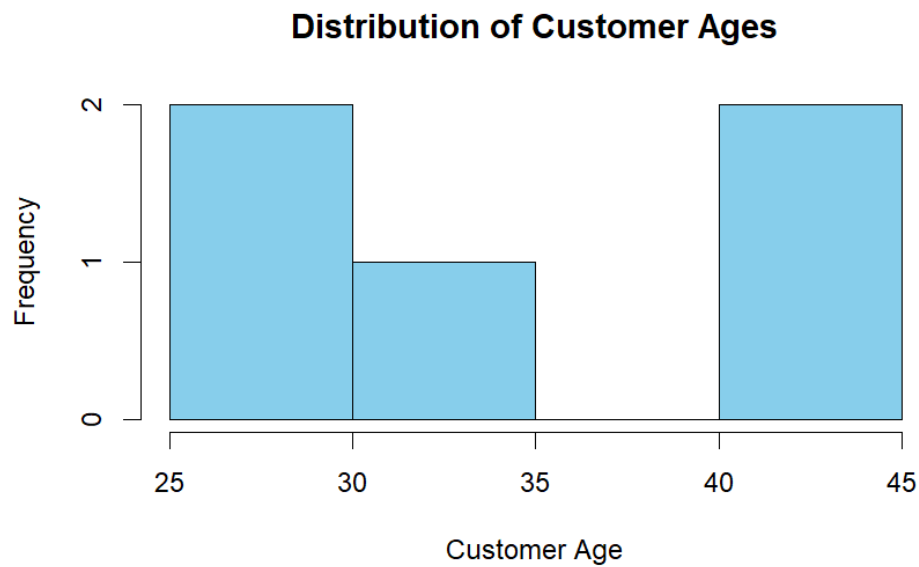


LAB EXPERIMENTS

SET-17

QUESTION-1



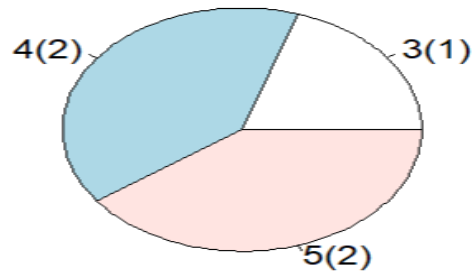
CODE:

```
# Customer age data
age <- c(28, 35, 42, 30, 45)

# Histogram
hist(age,
      col = "skyblue",
      main = "Distribution of Customer Ages",
      xlab = "Customer Age",
      ylab = "Frequency")
```

QUESTION-2

Customer Satisfaction Scores



CODE:

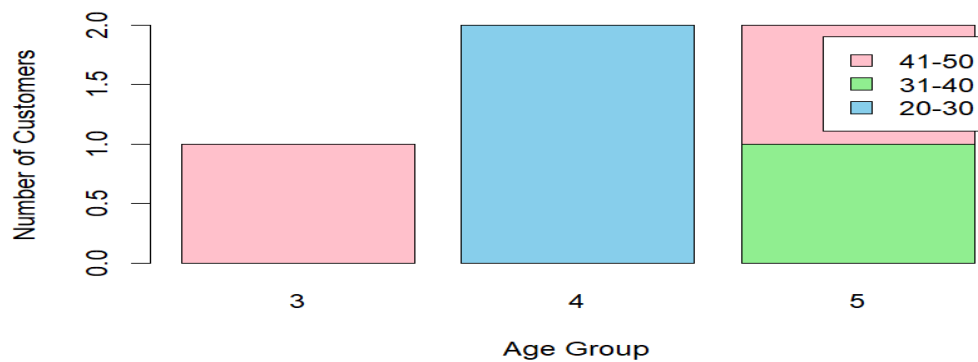
```
# Satisfaction scores
score <- c(4, 5, 3, 4, 5)

# Count each score
score_count <- table(score)

# Pie chart
pie(score_count,
    labels = paste(names(score_count),
        "(", score_count, ")", sep = "" ),
    main = "Customer Satisfaction Scores")
```

QUESTION-3

Customer Satisfaction by Age Group



CODE:

```
# Customer data
age <- c(28, 35, 42, 30, 45)
score <- c(4, 5, 3, 4, 5)

# Create age groups
age_group <- c("20-30", "31-40", "41-50", "20-30", "41-50")

# Create table
data <- table(age_group, score)

# Stacked bar chart
barplot(data,
  beside = FALSE,
  col = c("skyblue", "lightgreen", "pink"),
  main = "Customer Satisfaction by Age Group",
  xlab = "Age Group",
  ylab = "Number of Customers",
  legend.text = TRUE)
```

QUESTION-4



A word cloud visualization of customer feedback. The words are arranged in a circular pattern. The most prominent words in the center are 'excellent', 'customer', and 'delivery' in a large, bold, pink font. Surrounding these are other words in a smaller, green font, including 'friendly', 'bad', 'good', 'poor', and 'happy'. The words are oriented in various directions, creating a dynamic and visually engaging representation of the data.

CODE:

```

# Load packages

library(syuzhet)

library(wordcloud)

# Customer feedback

feedback <- c(

  "Good service",

  "Excellent customer support",

  "Fast delivery",

  "Very satisfied with the product",

  "Good quality product",

  "Poor delivery experience",

  "Excellent service",

  "Friendly staff",

  "Bad customer support",

  "Very happy with the purchase"

)

# Sentiment scores

sentiment <- get_sentiment(feedback, method = "syuzhet")

# Display sentiment scores

print(sentiment)

# Classify sentiment

for(i in 1:length(sentiment)) {

  if(sentiment[i] > 0) {

    print(paste(feedback[i], "-> Positive"))

  } else if(sentiment[i] < 0) {

    print(paste(feedback[i], "-> Negative"))

  } else {

    print(paste(feedback[i], "-> Neutral"))

  }

}

# Combine all feedback

```

```
text <- paste(feedback, collapse = " ")  
# Create word cloud  
wordcloud(words = text,  
           min.freq = 1,  
           random.order = FALSE,  
           colors = rainbow(10),  
           main = "Customer Feedback Word Cloud")
```